



Capstone: Retrieving, Processing, and Visualizing Data with Python

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Capstone: Retrieving, Processing, and Visualizing Data with Python

by University of Michigan

About this Course

In the capstone, students will build a series of applications to retrieve, process and visualize data using Python. The projects will involve all the elements of the specialization. In the first part of the capstone, students will do some visualizations to become familiar with the technologies in use and then will pursue their own project to visualize some other data that they have or can find. Chapters 15 and 16 from the book "Python for Everybody" will serve as the backbone for the capstone. This course covers Python 3.

Show less

Taught by:

Charles Russell Severance

Clinical Professor, School of Information

	Basic Info	Course 5 of 5 in the Python for Everybody Specialization
	Level	Beginner
	Commitment	6 weeks of study, 2-4 hours/week
	Language	English Subtitles: Arabic, French, Bengali, Uzbek, Ukrainian, Chinese (Simplified), Greek, Italian, Portuguese (Brazil), Vietnamese, Dutch, Korean, German, Pashto, Urdu, Russian, Thai, Indonesian, Swedish, Turkish, Azerbaijani, Spanish, Dari, Hindi, Japanese, Kazakh, Hungarian, Polish
	How To Pass	Pass all graded assignments to complete the course.
	User Ratings	★ 4.7

Syllabus

Module 1

Welcome to the Capstone

Congratulations to everyone for making it this far. Before you begin, please view the Introduction video and read the Capstone Overview. The Course Resources section contains additional course-wide material that you may want to refer to in future weeks.

4 videos, 5 readings

1. Video: Introduction: Welcome to the Class

2. Reading: Capstone Overview

3. Reading: Help Us Learn More About You!

4. Reading: Course Syllabus

5. Reading: Python Textbook

6. Video: Unicode Characters and Strings

7. Reading: Capstone Completion Options

8. Video: Office Hours in Den Haag, Netherlands

9. Video: Interview: John Resig and Pam Fox - Khan Academy

Graded: Python for Everybody - A Review

View Less

Module 2

Building a Search Engine

This week we will download and run a simple version of the Google PageRank Algorithm and practice spidering some content. The assignment is peer-graded, and the first of three optional Honors assignments in the course. This a continuation of the material covered in Course 4 of the specialization, and is based on Chapter 16 of the textbook.

6 videos, 2 readings

1. Reading: Building a Search Engine - Introduction

2. Video: Page Rank Overview

3. Video: Worked Example: Page Rank - Spidering

4. Video: Worked Example: Page Rank - Computation

5. Video: Worked Example: Page Rank - Visualization

6. Video: Office Hours Detroit, Michigan

7. Video: Interview: Anil Jain - Image Processing

8. Reading: Peer Graded Assignment - Instructor Input

Graded: Peer Grade: Page Rank

Module 3

Exploring Data Sources (Project)

The optional Capstone project is your opportunity to select, process, and visualize the data of your choice, and receive feedback from your peers. The project is not graded, and can be as simple or complex as you like. This week's assignment is to identify a data source and make a short discussion forum post describing the data source and outlining some possible analysis that could be done with it. You will not be required to use the data source presented here for your actual analysis.

2 videos, 2 readings

1. Reading: Identifying Your Data Source - Introduction

2. Reading: List of Data Sources (Instructional Staff Curated)

3. Discussion Prompt: Identifying a Data Source

4. Video: Dr. Chuck's New Kitten - Sakaiger

5. Video: Interview: Bruce Schneier - The Security Mindset

Module 4

Spidering and Modeling Email Data

In our second optional Honors assignment, we will retrieve and process email data from the Sakai open source project. Video lectures will walk you through the process of retrieving, cleaning up, and modeling the data.

5 videos, 1 reading

1. Reading: Spidering and Modeling Email Data - Introduction

2. Video: Gmane Introduction

3. Video: Worked Example: Gmane / Mail - Retrieval

4. Video: Worked Example: Gmane / Mail - Model

5. Video: Office Hours Baltimore, MD

6. Video: Interview: Bruce Schneier - Building Cryptographic Systems

Graded: Loading and Modeling Mail Data

Module 5

Accessing New Data Sources (Project)

The task for this week is to make a discussion thread post that reflects the progress you have made to date in retrieving and cleaning up your data source so can perform your analysis. Feedback from other students is encouraged to help you refine the process.

1 video, 1 reading

1. Reading: Accessing New Data Sources - Introduction

2. Discussion Prompt: Analyzing a Data Source

3. Video: Office Hours: Dr. Chuck Pretends to be Anthony Bourdain

Module 6

Visualizing Email Data

In the final optional Honors assignment, we will do two visualizations of the email data you have retrieved and processed: a word cloud to visualize the frequency distribution and a timeline to show how the data is changing over time.

3 videos, 1 reading

1. Reading: Visualizing Email Data

2. Video: Worked Example: Gmane / Mail - Visualization

3. Video: Office Hours, Montreal, Canada

4. Video: Interview: Nathaniel Borenstein - The Father of MIME

Graded: Visualizing Email Data

Module 7

Visualizing new Data Sources (Project)

This week you will discuss the analysis of your data to the class. While many of the projects will result in a visualization of the data, any other results of analyzing the data are equally valued, so use whatever form of analysis and display is most appropriate to the data set you have selected.

2 videos, 3 readings

1. Reading: Visualizing new Data Sources - Introduction

2. Discussion Prompt: Data Analysis and Visualization

3. Video: Office Hours - Dr. Chuck's Office - Ann Arbor, Michigan

4. Video: Video: Steve Jobs, NeXT and the Internet

5. Reading: Post-Course Survey

6. Reading: A special reason to celebrate your success

How It Works

General

What do start dates and end dates mean?

Once you enroll, you'll have access to all videos, readings, quizzes, and programming assignments (if applicable). If you choose to explore the content without purchasing, you may not be able to access certain assignments. If you don't finish all graded assignments before the end of the session, you can reset your deadlines. Your progress will be saved and you'll be able to pick up where you left off.

What are due dates? Is there a penalty for submitting my work after a due date?

Within a course, there are suggested due dates to help you manage your schedule and keep work from piling up. Quizzes and programming assignments can be submitted late without consequence. However, it is possible that you won't receive a grade if you submit your peer-graded assignment too late because classmates usually review assignment within three days of the assignment deadline.

Can I re-attempt an assignment?

Yes. If you want to improve your grade, you can always try again. If you're re-attempting a peer-graded assignment, re-submit your work as soon as you can to make sure there's enough time for your classmates to review your work. In some cases you may need to wait before re-submitting a programming assignment or quiz. We encourage you to review learning material during this delay.

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Course 5 of Specialization

Learn to Program and Analyze Data with Python

Develop programs to gather, clean, analyze, and visualize data.

University of Michigan

Python for Everybody

View the course in catalog